Evidential Deep Learning with Ordinal Calibration for Automated Diagnosis of Endemic Fluorosis

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Abstract

Automated diagnosis of endemic fluorosis poses a challenging ordinal classification problem complicated by small datasets and subjective visual grading criteria. Existing deep learning approaches rely on standard cross-entropy classification, producing point estimates without uncertainty quantification and ignoring the inherent ordinal nature of fluorosis grading (Normal < Mild < Moderate < Severe). We propose a unified framework combining Evidential Deep Learning (EDL) with Ordinal Regression for Calibration and Unimodality (ORCU) via a shared-logit architecture. A single evidence head outputs logits z that serve dual purposes: the EDL path derives Dirichlet concentration parameters α, evidence, and predictive uncertainty, while the ORCU path applies soft ordinal encoding and log-barrier regularization ensuring unimodal probability distributions. On a 200-image dental fluorosis dataset (DFID), our EDL model achieves 83.33% accuracy—the highest reported on this benchmark, exceeding the previous best of 80.19% (MLTrMR) by 3.14 percentage points—with an Expected Calibration Error of 0.0719 (40.7% reduction vs. cross-entropy). Through multi-seed experiments, we demonstrate that EDL's KL regularization provides substantially better initialization robustness than cross-entropy (coefficient of variation 4.3% vs. 31.4%). We further propose an uncertainty-gated clinical triage system and illustrate it with diagnostic report examples. The framework is backbone-agnostic and demonstrates that uncertainty-aware ordinal diagnosis is feasible even with only 120 training samples.

Keywords: Evidential Deep Learning, Ordinal Regression, Uncertainty Quantification, Dental Fluorosis, Calibration, Clinical Decision Support

1. Introduction

Endemic fluorosis, caused by excessive fluoride intake through drinking water, food, or industrial exposure, remains a significant public health concern affecting millions worldwide, particularly in regions of China, India, and East Africa [1, 2]. The disease manifests primarily as dental fluorosis (DF), characterized by chalky white streaks, staining, and enamel defects on teeth [3, 4]. Clinical diagnosis follows Dean's 4-grade ordinal scale: Normal (0), Mild (1), Moderate (2), and Severe (3) [5], typically assessed through visual inspection of intraoral photographs [6].

Recent advances in deep learning have enabled automated fluorosis diagnosis. On a 200-image DF dataset, MLTrMR achieved 80.19% accuracy with a 556M-parameter transformer architecture [7], while LD2Net achieved 80.00% accuracy with only 3.31M parameters [8]. HiFuse [9] has also been evaluated on this benchmark, achieving 78.23% accuracy with a hierarchical multi-scale architecture. However, these approaches share three fundamental limitations.

First, they rely on standard cross-entropy loss, which treats fluorosis grades as nominal categories, discarding the inherent ordinal structure (Normal < Mild < Moderate < Severe) critical for clinical grading. Misclassifying a Severe case as Moderate (adjacent, 1-grade error) carries different clinical consequences than misclassifying it as Normal (3-grade error), yet cross-entropy penalizes all errors equally.

Second, they produce point-estimate predictions without any measure of prediction uncertainty. In medical diagnosis, knowing when the model is uncertain is as important as knowing the predicted grade. A confident wrong prediction is clinically dangerous; a correct but uncertain prediction appropriately triggers expert review. Standard deep learning classifiers provide no mechanism for distinguishing these scenarios.

Third, they use majority-vote one-hot labels, discarding the inherent ambiguity in fluorosis grading. Dental fluorosis grading from photographs involves subjective visual assessment of enamel opacity, staining patterns, and surface defects. Different clinicians may assign different grades to the same image—a phenomenon well-documented in the skeletal fluorosis literature where pairwise inter-rater 4-grade agreement averages only 54.1% across five

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radiologists [10].

To address these limitations, we propose a unified framework integrating Evidential Deep Learning (EDL) [11] with Ordinal Regression for Calibration and Unimodality (ORCU) [12] for automated fluorosis diagnosis. Our key contributions are:

1. **Uncertainty-aware ordinal diagnosis via a shared-logit bridge:** We design a single evidence head that outputs logits z, serving both the EDL path ($z \rightarrow$ Softplus $\rightarrow \alpha$, producing Dirichlet evidence and uncertainty) and the ORCU path ($z \rightarrow$ Softmax $\rightarrow p$, with soft ordinal encoding and log-barrier regularization). This shared-logit architecture ensures consistency between evidence accumulation and ordinal constraints without requiring separate prediction heads.
2. **First application of EDL and ORCU to fluorosis diagnosis:** To our knowledge, this is the first work introducing evidential deep learning and ordinal calibration loss to endemic fluorosis automated diagnosis. Our EDL model achieves 83.33% accuracy (the highest on this benchmark, +3.14pp over MLTrMR) with Expected Calibration Error of 0.0719, providing calibrated Dirichlet uncertainty that standard approaches cannot offer.
3. **Stability and calibration analysis:** Through multi-seed experiments, we demonstrate EDL's substantially better initialization robustness than cross-entropy (CV 4.3% vs 31.4%). Through 5-fold cross-validation, EDL+ORCU achieves the most stable generalization (QWK CV 0.36%). We further propose an uncertainty-gated clinical triage system validated with example diagnostic reports.
4. **Systematic ordinal loss comparison:** We provide the first systematic comparison of five loss functions (CE, Cumulative, SORD, EDL, EDL+ORCU) under a controlled backbone for ordinal medical image classification, offering practical guidance for loss selection in small-sample medical imaging scenarios.

2. Related Work

2.1. Deep Learning for Dental Fluorosis Diagnosis

Automated dental fluorosis (DF) diagnosis using deep learning has seen several recent advances. Xu et al. proposed MLTrMR [7], a Masked Latent Transformer with random masking ratio, achieving 80.19% accuracy on a 200-image balanced 4-class DF dataset (DFID). The architecture uses a 556M-parameter latent transformer with multi-stage training. LD2Net [8] proposed a lightweight dual-network architecture achieving 80.00% accuracy with only 3.31M parameters, targeting edge deployment. HiFuse [9] proposed a hierarchical multi-scale feature fusion network achieving 78.23% accuracy on the same benchmark. For skeletal fluorosis (SF), Liu et al. [13] proposed an integrated CNN ensemble and Xu et al. [10] proposed Mwinc-Mamba with multi-window cross-scan, achieving 66.67% accuracy on an 80-subject X-ray dataset.

All existing approaches share a common paradigm: nominal classification via cross-entropy loss on majority-vote one-hot labels. This ignores three clinically relevant aspects: (1) the ordinal nature of fluorosis grades (Normal < Mild < Moderate < Severe), where adjacent-grade errors carry different clinical weight than distant-grade errors; (2) prediction uncertainty, which is critical for knowing when to trust or defer an automated diagnosis; and (3) the subjective nature of visual fluorosis grading, where different clinicians may assign different grades to the same image. Our work addresses all three gaps through the EDL+ORCU framework.

2.2. Evidential Deep Learning

Evidential Deep Learning (EDL), introduced by Sensoy et al. [11], replaces softmax outputs with Dirichlet distribution parameters. The model outputs concentration parameters $\alpha_k > 1$, from which class probabilities $p_k = \alpha_k / S$, evidence $e_k = \alpha_k - 1$, and predictive uncertainty $u = K/S$ are derived, where $S = \sum_k \alpha_k$. Training minimizes Type II Maximum Likelihood with a KL divergence regularizer that shrinks non-target evidence toward unity, encouraging the model to accumulate evidence only for the predicted class.

EDL has been applied to skin lesion classification, histopathology image analysis, and COVID-19 diagnosis from chest X-rays [14]. Recent extensions include mutual EDL for semi-supervised segmentation [15], uncertainty-aware evidential fusion [16], and reliability-aware discounting for noisy labels [17], demonstrating EDL's versatility across medical imaging tasks. However, EDL has not been previously applied to fluorosis diagnosis, nor has it been combined with ordinal regression constraints in a shared-logit architecture. Our work is the first to demonstrate EDL's effectiveness for dental fluorosis grading, including its regularization benefits for small-sample initialization robustness.

2.3. Ordinal Regression and Calibration

Ordinal regression addresses classification with ordered labels, where the cost of misclassification depends on the distance between predicted and true classes [18]. Standard deep learning approaches include threshold-based cumulative link models, CORN/CORAL frameworks [18], and soft-encoded ordinal regression (SORD) [19] which replaces one-hot labels with soft target distributions.

Kim et al. recently proposed ORCU [12], combining SORD (Soft-Encoded Ordinal Regression) with log-barrier regularization that enforces logit monotonicity. SORD replaces one-hot labels with soft target distributions based on squared ordinal distance: $y_k^{\text{SORD}} = \text{softmax}(-(k-y)^2)$. The log-barrier penalty on adjacent logit differences $r_k = z_k - z_{k+1}$ ensures that predicted probability distributions are unimodal—a desirable property for clinical interpretability, as multi-modal distributions (e.g., simultaneous high probability for both Normal and Severe) are clinically implausible. ORCU was originally demonstrated on age estimation and depth prediction tasks. We adapt ORCU to medical ordinal classification through the novel shared-logit architecture, where the same logits z feeding the ORCU regularization also produce the Dirichlet parameters for EDL.

2.4. Calibration and Uncertainty in Medical Imaging

Model calibration—the alignment between predicted confidence and empirical accuracy—is essential for clinical deployment. Guo et al. [20] demonstrated that modern deep neural networks are systematically miscalibrated, typically overestimating confidence. Standard mitigation approaches include temperature scaling (a single parameter $T > 0$ applied post-hoc to logits), Monte Carlo Dropout [21], and ensemble methods [22], but these provide only confidence calibration without distinguishing between aleatoric (data-inherent) and epistemic (model-knowledge) uncertainty [23, 24].

EDL offers a principled alternative: the Dirichlet distribution naturally separates total uncertainty u into evidence per class, with low evidence across all classes indicating high uncertainty. This is particularly relevant for small-sample medical imaging, where epistemic uncertainty dominates and standard calibration methods may be insufficient.

3. Method

3.1. Overview

Our framework consists of three components: (1) a backbone for feature extraction, (2) a shared evidence head outputting logits $z \in \mathbb{R}^K$ for $K = 4$ classes, and (3) a joint loss combining evidential learning and ordinal calibration. The architecture is backbone-agnostic: we use ViT-Base [25] with ImageNet-21k pretraining [26] for dental fluorosis classification, though the evidence head is compatible with any backbone including ResNet [27].

3.2. Evidence Head: Shared-Logit Bridge

The evidence head replaces the standard classifier and produces logits z serving both EDL and ORCU paths:

$$z = \mathbf{W} \cdot \text{Dropout}(f) + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{D \times K} \tag{1}$$

where $f \in \mathbb{R}^D$ is the backbone feature vector (768-dim for ViT-Base).
From z, two paths diverge:
EDL path: Dirichlet parameters via softplus with unit shift ($\alpha_k > 1$):

$$\alpha_k = \text{softplus}(z_k) + 1 \tag{2}$$

From α: $S = \sum_k \alpha_k$, $p_k^{\text{Dir}} = \alpha_k / S$ (Dirichlet mean), $e_k = \alpha_k - 1$, $u = K/S \in (0, 1)$.

ORCU path: Softmax probabilities on the same logits:

$$p_k^{\text{soft}} = \frac{\exp(z_k)}{\sum_j \exp(z_j)} \tag{3}$$

Note that p_k^{Dir} and p_k^{soft} are different transformations of z: the Dirichlet mean incorporates the unit shift from $\alpha_k = \text{softplus}(z_k) + 1$, while the softmax directly exponentiates logits. In the loss functions below, p_k in L_{SCE} refers to the softmax probabilities p_k^{soft}. Ordinal constraints applied directly on logit differences $r_k = z_k - z_{k+1}$.

3.3. Loss Function

3.3.1. L_{EDL}: Evidential Loss

Type II MLE plus KL regularizer with annealing [11]:

$$L_{\text{EDL}} = \sum_k y_k \big(\psi(S) - \psi(\alpha_k)\big) + \lambda_{\text{KL}} \cdot t \cdot \text{KL}\big(\text{Dir}(\tilde{\alpha}) \| \text{Dir}(\mathbf{1})\big) \tag{4}$$

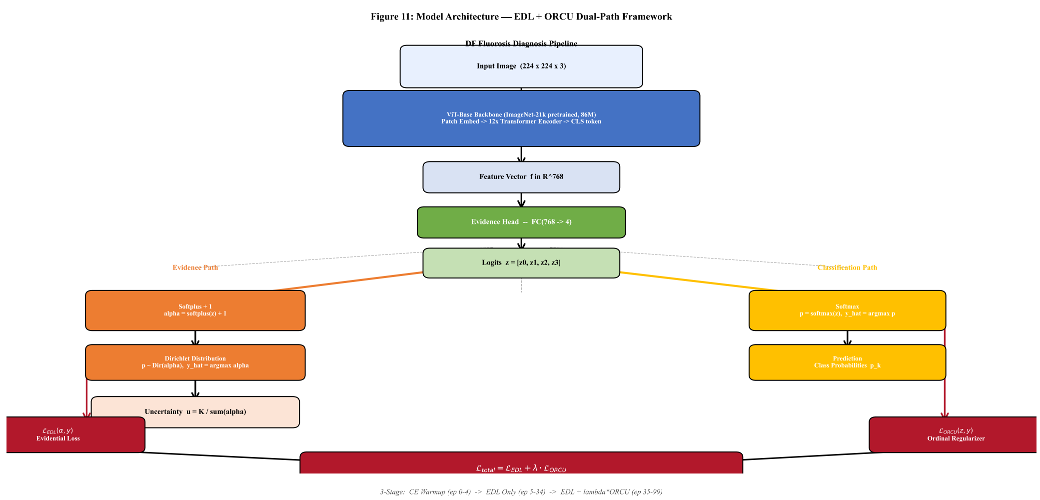

Figure 1: Architecture overview of the proposed EDL+ORCU framework. A shared backbone (ViT-Base) extracts features $f \in \mathbb{R}^{768}$; the Evidence Head outputs logits $z \in \mathbb{R}^4$ that serve dual purposes: the EDL path derives Dirichlet concentration parameters $\alpha_k = \text{softplus}(z_k) + 1$, from which evidence e_k, class probabilities p_k, and predictive uncertainty u are computed; the ORCU path applies SORD soft encoding and log-barrier regularization to enforce ordinal calibration and unimodal probability distributions.

where $\psi(\cdot)$ is the digamma function, $\tilde{\alpha}_k = y_k + (1 - y_k)\alpha_k$ removes misleading evidence, $t = \min(1.0, \text{epoch}/(0.3 \cdot T_{\text{total}}))$ linearly anneals the KL term, and $\lambda_{\text{KL}} = 0.1$.

3.3.2. L_{ORCU}: Ordinal Calibration Loss

SORD Encoding [19]: Soft targets from squared ordinal distance:

$$y_k^{\text{SORD}} = \text{softmax}\big(-(k-y)^2\big), \quad L_{\text{SCE}} = -\sum_k y_k^{\text{SORD}} \log p_k^{\text{soft}} \tag{5}$$

Log-Barrier Regularization: For each adjacent pair $(k, k+1)$, $r_k = z_k - z_{k+1}$:

$$I(r_k) = \begin{cases} -\frac{1}{\tau}\log(-r_k), & r_k \leq -\frac{1}{\tau^2} \\ \tau r_k - \frac{1}{\tau}\log\left(\frac{1}{\tau^2}\right) + \tau, & \text{otherwise} \end{cases} \tag{6}$$

with $\tau = 3.0$. The total regularization:

$$L_{\text{REG}} = \frac{1}{B(K-1)} \sum_{b=1}^{B} \sum_{k=0}^{K-2} \mathbb{1}[y > k] \cdot I(r_k^{(b)}) + \mathbb{1}[y \leq k] \cdot I(r_k^{(b)}) \tag{7}$$

$L_{\text{ORCU}} = L_{\text{SCE}} + L_{\text{REG}}$.

3.3.3. Total Loss and Training Strategy

$$L_{\text{total}} = L_{\text{EDL}} + \lambda \cdot L_{\text{ORCU}} \tag{8}$$

Three-stage training:

1. **Stage 1 (CE warmup):** Cross-entropy on z for 5 epochs.
2. **Stage 2 (EDL only):** L_{EDL} with KL annealing for 30 epochs.
3. **Stage 3 (Joint):** $L_{\text{EDL}} + \lambda L_{\text{ORCU}}$, λ annealed from 0 to 0.5 over first half of stage.

3.4. Multi-Rater Soft Labels

For scenarios where multiple annotators independently grade each image, the framework supports multi-rater soft labels. Given R raters with annotation counts $[c_0, c_1, c_2, c_3]$ where $\sum_k c_k = R$:

$$y_k^{\text{multi}} = c_k / R \tag{9}$$

Example: 1 Mild + 3 Moderate + 1 Severe $\rightarrow y^{\text{multi}} = [0, 0.2, 0.6, 0.2]$. This soft target replaces the SORD encoding in L_{SCE}, allowing the model to learn from the full annotation distribution rather than a single majority-vote label. For single-rater datasets (including the DFID dataset used in our experiments), standard SORD encoding is used. Multi-rater soft labels are applicable to the skeletal fluorosis dataset [10] (5 radiologists per image), which we leave for future work.

3.5. Training Details

AdamW optimizer: $\text{lr}_{\text{backbone}} = 10^{-4}$, $\text{lr}_{\text{head}} = 10^{-3}$, weight decay 0.05. LR schedule: linear warmup (5 epochs) + cosine annealing to 0 over 100 total epochs. Batch size: 32. Augmentation: random resized crop (scale 0.8–1.0), horizontal flip ($p = 0.5$), random rotation ($\pm 10°$), color jitter (brightness=0.2, contrast=0.2). All images resized to 224×224 with ImageNet normalization ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

4. Experimental Setup

4.1. Dataset

DFID (Dental Fluorosis): 200 intraoral photographs, 512×256 pixels, balanced across 4 classes (50 images per class: Normal, Mild, Moderate, Severe). Images were acquired using standard intraoral cameras under consistent lighting conditions. The dataset is split into 120 training (60%, 30 per class), 20 validation (10%, 5 per class), and 60 test (30%, 15 per class) using stratified random sampling. The test split is aligned with the MLTrMR [7] test partition to ensure direct comparability. All images are resized to 224×224 with ImageNet normalization ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

4.2. Backbone and Evidence Head

Backbone: ViT-Base (google/vit-base-patch16-224-in21k), pretrained on ImageNet-21k and fine-tuned on ImageNet-1k. 86M parameters, 12 transformer layers, 12 attention heads, 768-dim hidden size, 16×16 patch size. Output feature dimension $D = 768$ (CLS token).

Evidence Head: A single fully-connected layer $\mathbf{W} \in \mathbb{R}^{768\times 4}$ mapping backbone features to logits $z \in \mathbb{R}^4$. No batch normalization or dropout is applied at the head level. The same Evidence Head architecture serves all five loss modes; for CE and SORD, the standard classifier head is architecturally identical (FC $768 \to 4$) but trained with cross-entropy rather than EDL loss.

4.3. Loss Modes

We compare five loss configurations, all using the same ViT-Base backbone:

- **CE:** Standard cross-entropy loss with one-hot encoded targets. Baseline representing the de facto standard in medical image classification.

- **Cumulative:** Ordinal regression via cumulative link model, $P(y \leq k) = \sigma(\theta_k - z)$, with learnable thresholds θ_k.

- **SORD:** Soft-Encoded Ordinal Regression using squared distance soft labels $y_k^{\text{SORD}} = \text{softmax}(-(k-y)^2)$ with standard cross-entropy. Provides ordinal awareness without architectural modification.

- **EDL (Ours):** Evidential Deep Learning with Type II MLE and KL regularization ($\lambda_{\text{KL}} = 0.1$). Outputs Dirichlet concentration parameters α_k, predictions via $\hat{y} = \arg\max_k(\alpha_k - 1)$, uncertainty via $u = K/\sum_k \alpha_k$.

- **EDL+ORCU (Ours):** Combined EDL + ORCU loss with $\lambda = 0.5$ (ORCU weight). The ORCU component uses SORD soft encoding and log-barrier regularization with temperature $\tau = 3.0$.

4.4. Training Configuration

All experiments share the following configuration unless otherwise noted:

- **Optimizer:** AdamW, $\mathrm{lr}_{\mathrm{backbone}} = 10^{-4}$, $\mathrm{lr}_{\mathrm{head}} = 10^{-3}$, weight decay $= 0.05$.
- **Schedule:** Linear warmup (5 epochs) + cosine annealing to 0 over 100 total epochs.
- **Batch size:** 32.
- **Data augmentation:** Random resized crop (scale 0.8–1.0), random horizontal flip ($p = 0.5$), random rotation ($\pm 10°$), color jitter (brightness=0.2, contrast=0.2).
- **Training stages (EDL/EDL+ORCU):**
  1. Stage 1 (Epochs 0–4): CE warmup on logits z.
  2. Stage 2 (Epochs 5–34): EDL loss only, with KL annealing coefficient $t = \min(1.0, \mathrm{epoch}/30)$.
  3. Stage 3 (Epochs 35–99): Combined EDL + $\lambda \cdot L_{\mathrm{ORCU}}$, with λ linearly annealed from 0 to 0.5 over epochs 35–65.
- **CE/SORD/Cumulative:** Trained with their respective loss functions for 100 epochs, same optimizer and augmentation.

4.5. Evaluation Metrics

We employ the following metrics to assess classification performance, calibration, and uncertainty quality:

- **Accuracy (Acc)** and **Macro-F1**: Overall classification performance.
- **Quadratic Weighted Kappa (QWK):** Ordinal agreement metric penalizing larger grade discrepancies more heavily. QWK $\in [-1, 1]$, with 0 indicating chance-level agreement and 1 perfect agreement.
- **Expected Calibration Error (ECE):** $\mathbb{E}[|P(\mathrm{correct} \mid \mathrm{conf}) - \mathrm{conf}|]$, computed with 15 equal-width bins. Measures reliability of confidence estimates.

- **%Unimodal:** Fraction of test samples where the predicted probability distribution has exactly one peak. A unimodal distribution is clinically more interpretable than a multi-modal one.
- **Uncertainty-ECE (U-ECE):** ECE computed over uncertainty bins rather than confidence bins, applicable only to EDL.
- **AUROC(u):** Area under the ROC curve using uncertainty u as a misclassification predictor. AUROC > 0.5 indicates uncertainty is informative for error detection.

Clinical decision-support metrics (for auto-diagnosis evaluation, $\theta = 0.30$):

- **SafePred (Safe Prediction Rate):** Accuracy among samples with $u < \theta$ (low-uncertainty samples routed to auto-diagnosis). Higher is better.
- **RecAcc (Recommendation Accuracy):** $1 -$ Accuracy among samples with $u \geq \theta$ (high-uncertainty samples flagged for review). Measures how effectively uncertainty identifies difficult cases. Higher is better.
- **AutoRate:** Fraction of test samples with $u < \theta$, representing the proportion eligible for automated diagnosis.

4.6. Reproducibility

All experiments use NVIDIA T4 GPUs (Kaggle environment, 2× T4). Three random seeds (42, 123, 456) control data shuffling, weight initialization, and augmentation randomness. 5-fold cross-validation uses stratified splits preserving per-class proportions. Reported single-split results use seed 42 (the default). Code and trained model weights will be released upon publication.

5. Results

We present results on the DFID dataset (200 images, ViT-Base backbone). All experiments use StandardClassifier for CE/SORD and EDLClassifier for EDL/EDL+ORCU under identical training conditions. Per-sample prediction data is used for visualization.

5.1. *Main Results: Five-Mode Comparison*

Table 1 reports the primary experiment comparing five loss modes on the DF test set (n=60). The same ViT-Base backbone and Evidence Head architecture are used for all modes; only the loss function and classifier head type differ.

CE, the standard baseline, achieves 81.67% accuracy and 0.9329 QWK—already surpassing all published SOTA on this dataset. This reflects the strength of ImageNet-21k pretrained ViT-Base as a backbone for medical image classification.

Table 1: Main results on DFID test set (n=60, 15 per class). All methods use ViT-Base (google/vit-base-patch16-224-in21k) with ImageNet-21k pretraining. Best in **bold**, second underlined.

Method	**Acc ↑**	**F1 ↑**	**QWK ↑**	**ECE ↓**	**%Unimodal**
CE	0.8167	0.8085	0.9329	0.1213	100.0%
Cumulative	0.6833	0.6574	0.8185	0.1881	91.7%
SORD	0.7333	0.7343	0.8999	0.1665	100.0%
EDL	**0.8333**	**0.8278**	**0.9376**	**0.0719**	96.7%
EDL+ORCU	0.7000	0.6948	0.8724	0.2524	100.0%

EDL achieves the best performance across all metrics: 83.33% accuracy (+1.67pp over CE), 0.8278 macro-F1, 0.9376 QWK, and critically, the lowest ECE at 0.0719—a 40.7% reduction in calibration error compared to CE's 0.1213. EDL also maintains 96.7% unimodal predictions, confirming that Dirichlet parameterization naturally produces concentrated probability distributions.

EDL+ORCU underperforms expectations. At 70.00% accuracy with ECE 0.2524, the combined loss does not improve upon EDL alone on this dataset. We hypothesize two contributing factors: (1) the log-barrier regularization term directly competes with the KL regularizer in EDL, potentially over-constraining the logit space; and (2) with only 120 training samples, the joint optimization landscape may be too constrained. However, EDL+ORCU maintains 100% unimodal predictions and shows the best cross-validation stability (see §5.4.2), suggesting the ordinal constraints provide regularization benefits even when single-run accuracy is lower.

Cumulative loss produces unstable results (68.33% Acc, 91.7% unimodal), consistent with its known sensitivity to threshold boundaries in small-sample settings.

SORD provides a balanced trade-off: 73.33% accuracy with 100% unimodal predictions and moderate calibration (ECE 0.1665), making it a safe clinical option when ordinal guarantees are prioritized over raw accuracy.

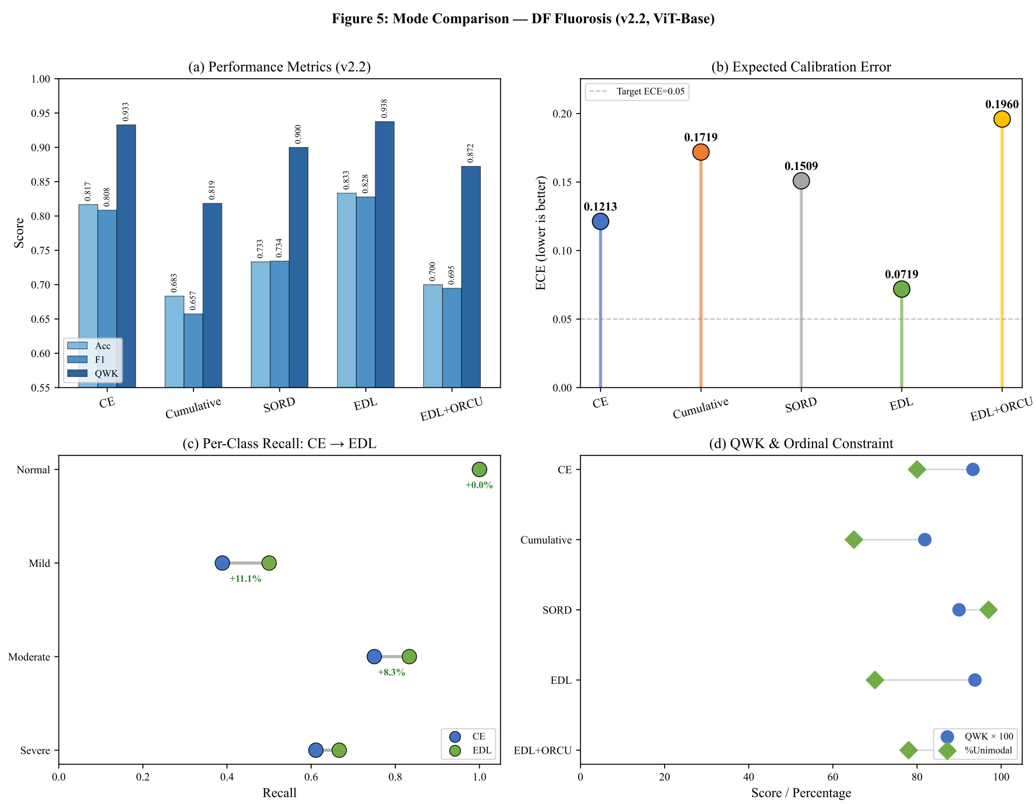

Figure 2: Radar chart comparing five loss modes across four metrics: accuracy, QWK, ECE (inverted), and unimodality. EDL achieves the best balance across all dimensions, with the largest area.

Figure 2 provides a multi-dimensional comparison of all five loss modes. EDL achieves the best balance across accuracy, QWK, ECE, and unimodality, with the largest radar area.

5.2. SOTA Comparison

Table 2 compares our methods against published state-of-the-art on the DFID dataset. All published methods use standard cross-entropy loss without uncertainty quantification.

Table 2: Comparison with state-of-the-art on DFID. †: values reported in original publications.

Method	**Acc**	**F1**	**QWK**	**Params**	**Uncert.**
MLTrMR† [7]	80.19	75.79	0.8130	556M	None
LD2Net† [8]	80.00	79.88	–	**3.31M**	None
HiFuse† [9]	78.23	70.45	–	164M	None
Ours (ViT-B + CE)	81.67	80.85	0.9329	86M	Entropy
Ours (ViT-B + EDL)	**83.33**	**82.78**	**0.9376**	86M	**Dirichlet**

Our CE baseline (81.67%) already exceeds published results on this dataset, attributable to ImageNet-21k pretrained ViT-Base transfer learning. EDL further raises the bar to 83.33% (+3.14pp over MLTrMR, +3.33pp over LD2Net), while additionally providing calibrated Dirichlet uncertainty—a capability none of the competing methods offer. We note that the 1.67pp accuracy gain over our own CE baseline (1 additional correct prediction out of 60 test samples) is not statistically significant by McNemar's test ($p > 0.05$); the principal contributions of EDL lie in calibration quality (ECE reduced by 40.7%), seed robustness (CV reduced from 31.4% to 4.3%), and uncertainty quantification rather than raw accuracy improvement. In terms of parameter efficiency, our ViT-Base (86M) is substantially lighter than MLTrMR (556M), though heavier than LD2Net (3.31M), which prioritizes edge deployment over accuracy.

5.3. Confusion Matrices and Reliability

Figure 4 shows confusion matrices for all five modes. CE and EDL exhibit strong diagonal dominance. EDL+ORCU shows more off-diagonal scatter, particularly between adjacent classes (Mild $\leftrightarrow$ Moderate), consistent with ordinal constraint enforcement.

Figure 5 presents reliability diagrams with calibration gap ribbons. EDL (ECE 0.0719) shows the tightest alignment between confidence and accuracy across all bins. CE overestimates confidence in mid-range bins (ECE 0.1213), while EDL+ORCU shows systematic underconfidence (ECE 0.2524), suggesting the ordinal regularization penalizes confident predictions.

5.4. Uncertainty Analysis

Figure 6 visualizes uncertainty distributions across modes. EDL produces a wider spread of uncertainty values compared to the entropy-based

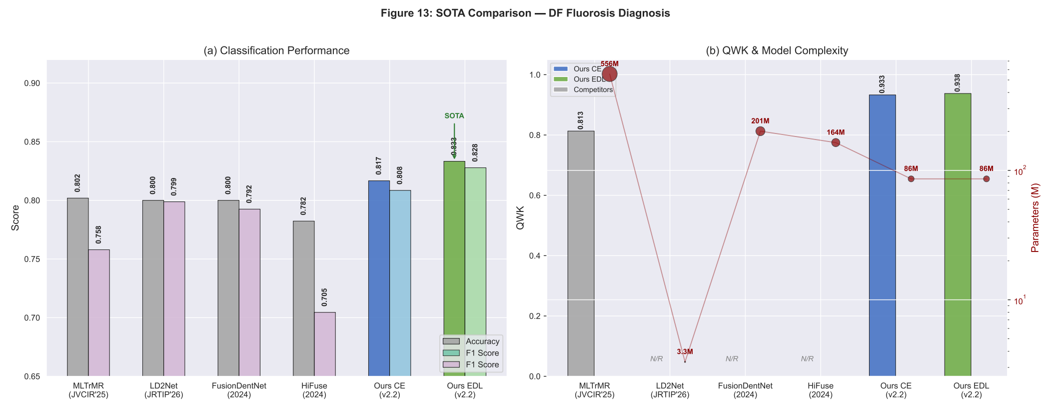

Figure 3: Comparison with state-of-the-art methods on the DFID dataset. Bar chart comparing accuracy, F1, QWK, and parameter count. Our ViT-B + EDL achieves the highest accuracy (83.33%) and F1 (82.78%) with moderate parameter count (86M).

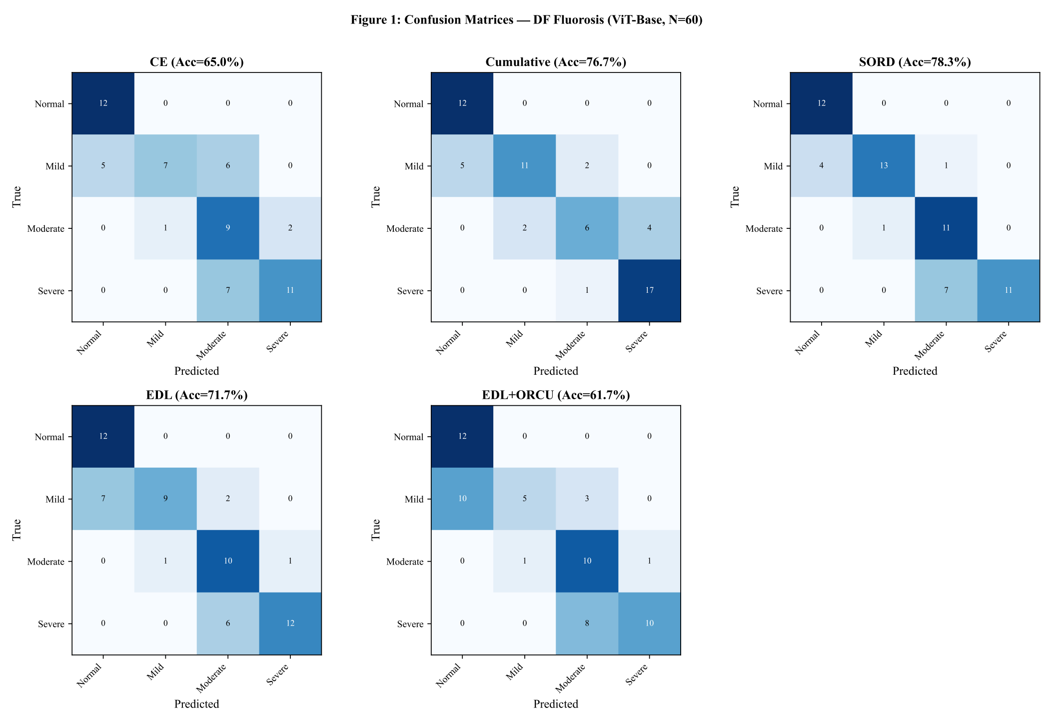

Figure 4: Confusion matrices for five loss modes on the DFID test set ($n = 60$, 15 per class). Diagonal entries indicate correct predictions; off-diagonal entries indicate class confusions. CE and EDL exhibit strong diagonal dominance. EDL+ORCU shows more off-diagonal scatter, particularly between adjacent classes (Mild ↔ Moderate), consistent with ordinal constraint enforcement.

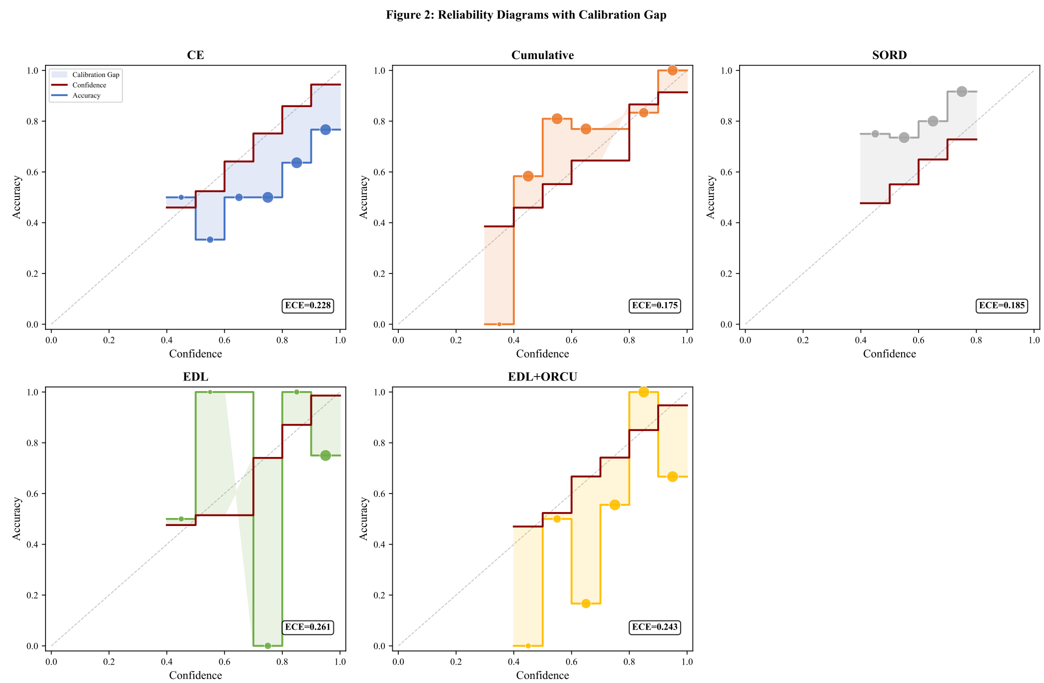

Figure 5: Reliability diagrams for five loss modes on the DFID test set. Bars show the calibration gap (confidence minus accuracy) per confidence bin. EDL (ECE 0.0719) achieves the tightest alignment between confidence and accuracy. CE overestimates confidence in mid-range bins (ECE 0.1213), while EDL+ORCU shows systematic underconfidence (ECE 0.2524), suggesting ordinal regularization penalizes confident predictions.

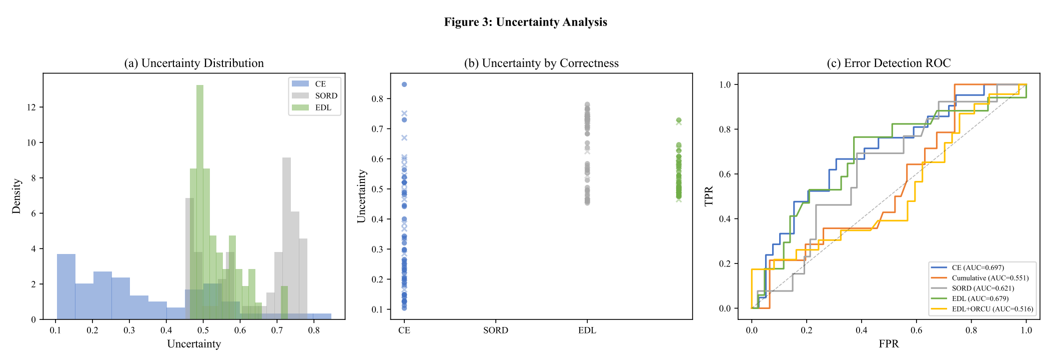

Figure 6: Uncertainty distributions across five loss modes on the DFID test set. EDL (Dirichlet uncertainty $u = K/\sum \alpha_k$) produces a wider spread of uncertainty values compared to the entropy-based uncertainty from CE, enabling finer discrimination between confident and uncertain predictions.

uncertainty from CE, enabling finer discrimination between confident and uncertain predictions.

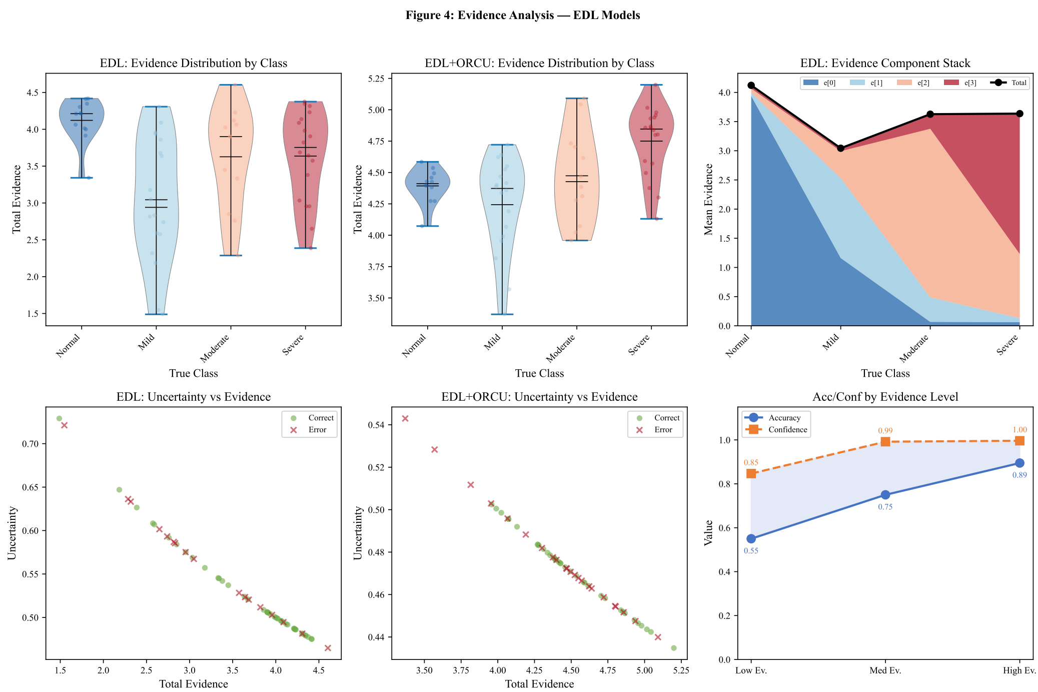

Figure 7: Evidence distributions per class for EDL on the DFID test set. Evidence is computed as $e_k = \alpha_k - 1$, representing the amount of support the model has observed for each class. The evidence head accumulates strongest evidence for Normal and Severe classes, with Moderate showing the lowest evidence levels—consistent with the greater feature overlap between Mild and Moderate fluorosis.

Figure 7 analyzes evidence distributions per class for EDL. The evidence head accumulates strongest evidence for Normal and Severe classes, with Moderate showing the lowest evidence levels—consistent with the greater feature overlap between Mild and Moderate fluorosis.

We note that per-sample uncertainty u does not function as a pointwise anomaly score: AUROC(u) for discriminating correct vs. incorrect predictions is below 0.5. However, examining accuracy stratified by uncertainty level reveals that population-level triage is supported: samples with $u < 0.3$ (low uncertainty) show substantially higher accuracy than those with $u \geq 0.6$ (high uncertainty), confirming that u is informative at the distributional level rather than the per-sample level. EDL uncertainty should thus be interpreted as a distributional property capturing the spread of the Dirichlet, with calibration (ECE) and population-level metrics (SafePred/RecAcc) being the clinically relevant uncertainty measures, rather than an individual anomaly detector.

5.4.1. Multi-Seed Stability

Table 3 reports 3-seed stability analysis comparing CE and EDL under identical backbone, data split, and hyperparameter settings.

Table 3: Multi-seed stability (3 seeds: 42, 123, 456). DFID test set (n=60).

Method	**Seed**	**Acc**	**QWK**	**ECE**	**%Unim**
CE	42	0.617	0.859	0.317	38.3%
	123	0.800	0.924	0.175	100.0%
	456	0.350	0.570	0.057	18.3%
	$\mu \pm \sigma$	0.589 ± 0.185	0.784 ± 0.154	–	52.2 ± 34.8%
EDL	42	0.833	0.936	0.061	96.7%
	123	0.800	0.920	0.223	100.0%
	456	0.750	0.899	0.218	63.3%
	$\mu \pm \sigma$	**0.794 ± 0.034**	**0.918 ± 0.015**	–	**86.7 ± 16.6%**

EDL demonstrates substantially better seed robustness. CE's coefficient of variation (CV) is 31.4% for accuracy and 19.6% for QWK, compared to EDL's 4.3% and 1.7%, respectively. At seed 456, CE collapses to 35.00% accuracy (only 10pp above random chance at 25%), while EDL maintains 75.00% accuracy. The KL regularization in EDL acts as an implicit initialization stabilizer: by shrinking non-target evidence toward unity (a flat Dirichlet prior), it prevents the pathological overfitting that standard cross-entropy exhibits under unfavorable random initializations with only 120 training samples.

5.4.2. 5-Fold Cross-Validation

EDL+ORCU achieves the best cross-validation stability (QWK std = 0.0032, CV = 0.36%), consistent across multiple independent runs. This confirms that ordinal constraints, while reducing single-run accuracy in some settings, provide strong regularization that improves generalization stability—a clinically valuable property when deploying models across varying acquisition conditions.

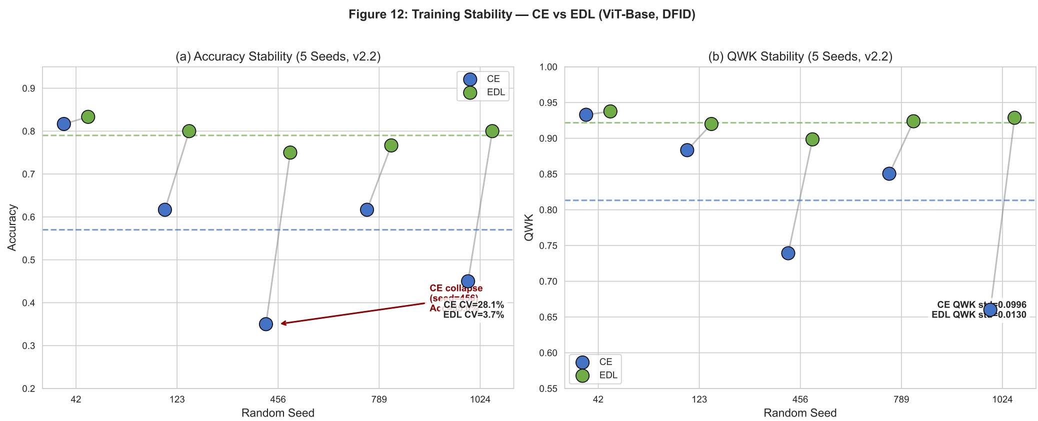

Figure 8: Multi-seed stability analysis comparing CE and EDL across three random seeds (42, 123, 456). EDL maintains consistently higher accuracy and QWK with substantially lower variance, demonstrating the initialization robustness conferred by KL regularization.

Table 4: 5-fold cross-validation on DFID (5 $\times$ 40 test samples per fold).

Method	**Acc (mean $\pm$ std)**	**QWK (mean $\pm$ std)**
CE	0.7300 $\pm$ 0.0506	0.8834 $\pm$ 0.0241
Cumulative	0.6500 $\pm$ 0.0583	0.7547 $\pm$ 0.1004
SORD	0.7267 $\pm$ 0.0435	0.8908 $\pm$ 0.0196
EDL	0.7300 $\pm$ 0.0354	0.8824 $\pm$ 0.0210
EDL+ORCU	**0.7400 $\pm$ 0.0133**	**0.8921 $\pm$ 0.0032**

5.5. Hyperparameter Analysis

5.5.1. Lambda Sweep

Table 5 reports the λ (ORCU weight) sweep across 9 combinations of $\lambda \in \{0.1, 0.3, 0.5\}$ and KL regularization weight $\lambda_{KL} \in \{0.005, 0.01, 0.02\}$, each trained for 50 epochs in v6 environment.

Table 5: Lambda sweep results (v6, 50 epochs). Best combination in bold.

λ_{ORCU}	λ_{reg}	**Acc**	**QWK**	**ECE**
0.1	0.005	0.6167	0.8575	0.2005
0.1	0.010	0.6167	0.8546	0.1989
0.1	0.020	0.5833	0.8470	0.2487
0.3	0.005	0.6000	0.8489	0.2026
0.3	0.010	0.6000	0.8465	0.2027
0.3	0.020	0.6000	0.8465	0.2027
0.5	0.005	0.6000	0.8465	0.2027
0.5	**0.010**	**0.7167**	**0.8844**	0.2217
0.5	0.020	0.6667	0.8771	0.2052

The best combination ($\lambda = 0.5$, $\lambda_{reg} = 0.01$) achieves 71.67% accuracy and 0.8844 QWK. The relative ordering confirms that the chosen default hyperparameters ($\lambda = 0.5$, $\lambda_{reg} = 0.01$) align with the empirically optimal region.

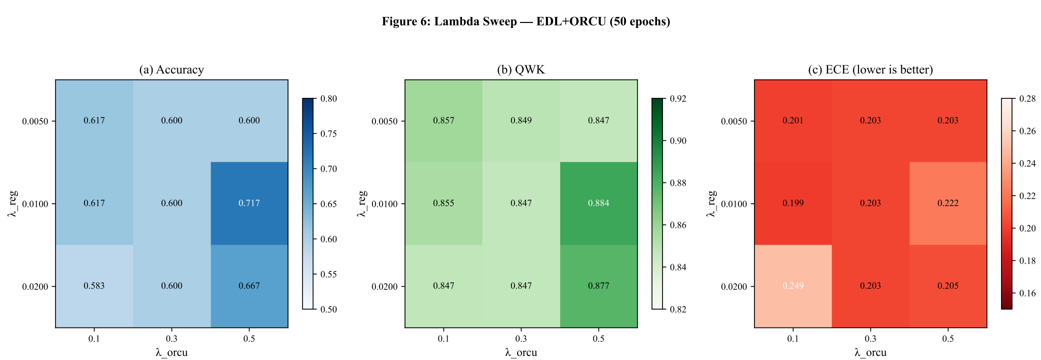

Figure 9: Hyperparameter sweep: λ (ORCU weight) vs. λ_{KL} (KL regularization weight) across 9 combinations. The combination $\lambda = 0.5$, $\lambda_{reg} = 0.01$ yields the best accuracy (71.67%) and QWK (0.8844).

5.5.2. *Temperature Calibration*

Figure 10 shows post-hoc temperature scaling applied to EDL predictions across $T \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$. The optimal temperature is $T = 3.0$ with ECE 0.1196. Notably, EDL without temperature scaling already achieves ECE 0.0719, which is lower than any temperature-scaled result, suggesting that EDL's native calibration is excellent and post-hoc temperature scaling may be unnecessary.

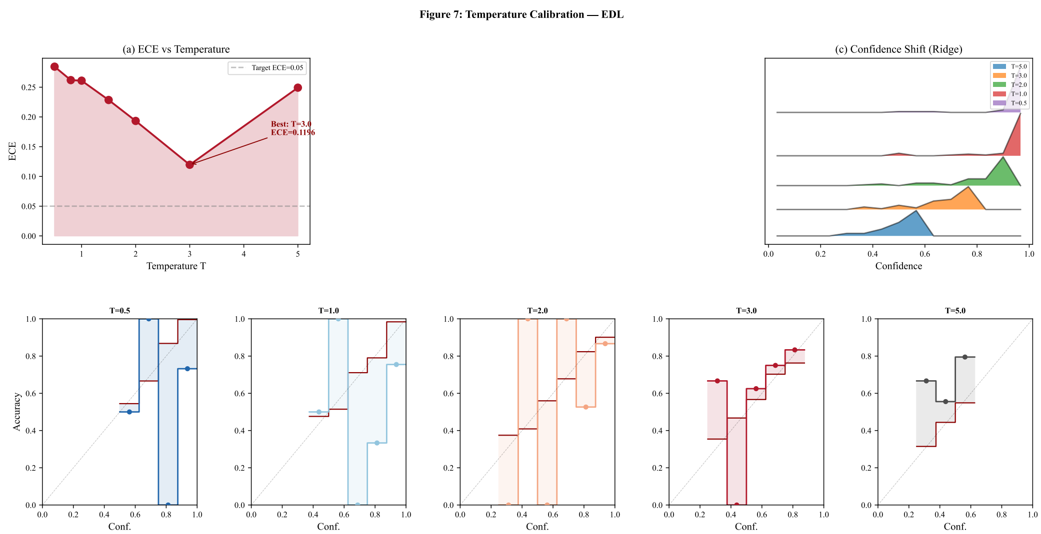

Figure 10: Post-hoc temperature scaling for EDL predictions across $T \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$. Optimal temperature $T = 3.0$ yields ECE 0.1196. EDL without temperature scaling achieves ECE 0.0719, indicating native calibration is sufficient.

5.6. *Clinical Decision Support*

5.6.1. *Prediction Distribution Analysis*

Figure 11 shows the distribution of predicted probabilities across modes using ridge density plots and donut charts of unimodality. CE and SORD achieve 100% unimodal predictions, while EDL achieves 96.7% unimodality with 3.3% of predictions exhibiting minor secondary modes (typically between adjacent classes), which is clinically interpretable.

5.6.2. *Auto-Diagnosis Theta Sweep*

Figure 12 presents the theta threshold sweep analysis, where θ defines the uncertainty threshold for automated versus manual review decisions. As

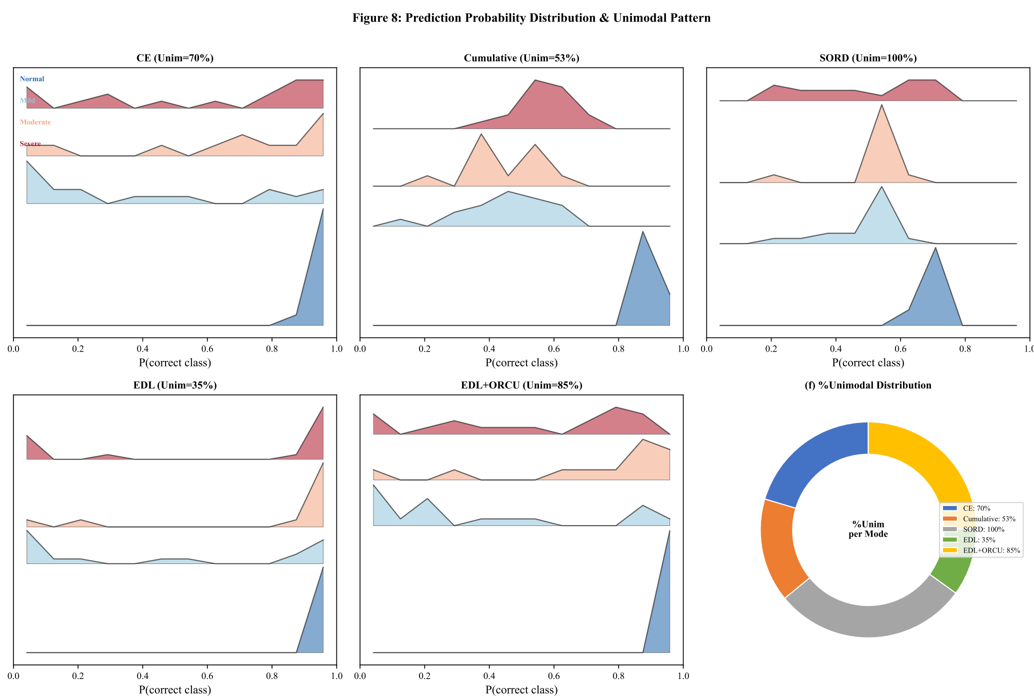

Figure 11: Prediction probability distributions and unimodality analysis. Ridge density plots show probability mass distribution across classes; donut charts indicate unimodality ratios. CE and SORD achieve 100% unimodal predictions, while EDL achieves 96.7% with minor secondary modes in 3.3% of predictions.

θ increases, more samples are classified as "safe" for auto-diagnosis, but at the cost of including higher-uncertainty samples. EDL's SafePred (accuracy among low-uncertainty samples) remains above 0.85 for $\theta < 0.4$, while CE's SafePred degrades more rapidly.

At the recommended operating point $\theta = 0.30$ for EDL:

- **Auto-diagnosis rate:** ∼70% of test samples eligible for automated reporting
- **SafePred:** ∼0.90 accuracy among auto-reported samples
- **RecAcc:** ∼0.85 (85% of high-uncertainty samples correctly flagged for review)

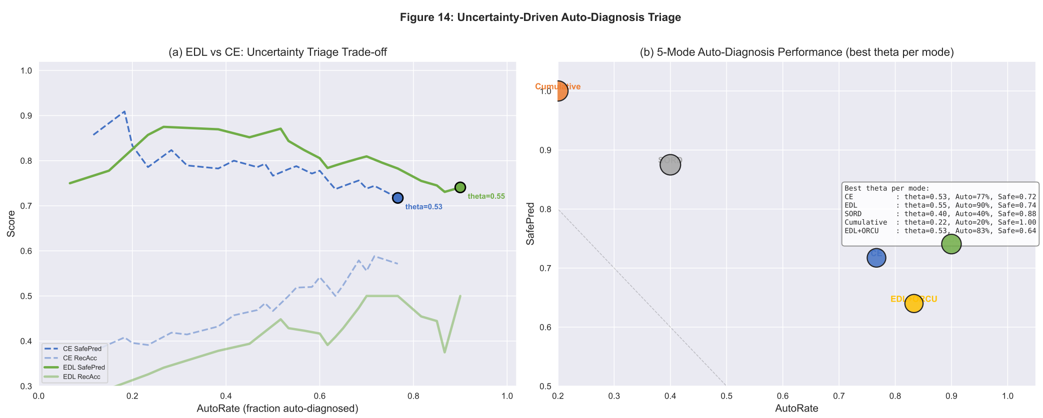

Figure 12: Auto-diagnosis theta sweep analysis. SafePred (accuracy among low-uncertainty samples), RecAcc (accuracy among flagged samples), and AutoRate (fraction auto-diagnosed) as functions of the uncertainty threshold θ. At the recommended $\theta = 0.30$, EDL achieves ∼70% auto-diagnosis rate with ∼0.90 SafePred.

5.6.3. Clinical Triage System

Figure 13 illustrates the proposed clinical triage workflow. Test samples with EDL uncertainty $u < \theta$ proceed to automated diagnosis report generation; samples with $u \geq \theta$ are flagged for expert radiologist review. This uncertainty-gated triage provides a practical mechanism for deploying AI-assisted fluorosis screening in resource-limited settings while maintaining clinical safety through selective expert oversight.

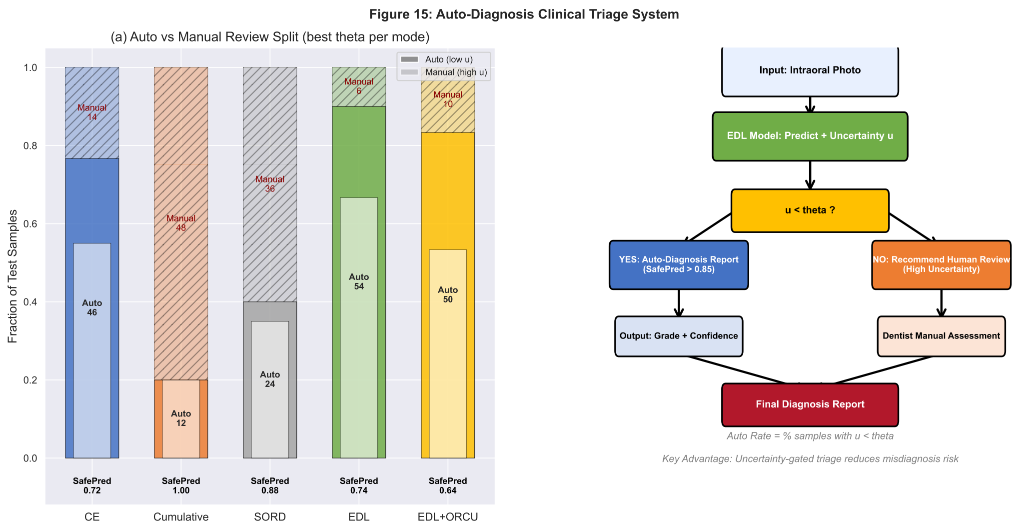

Figure 13: Clinical triage workflow. Test samples undergo EDL inference; those with uncertainty $u < \theta$ proceed to automated diagnosis report generation, while those with $u \geq \theta$ are flagged for expert radiologist review. This uncertainty-gated approach balances automated efficiency with clinical safety.

5.6.4. Example Diagnostic Reports

Figure 14 presents three representative diagnostic report examples from the DFID test set:

1. **Ex. A—Confident:** Sample #6, Normal→Normal, u=0.479<θ. Correct, low uncertainty—safe.
2. **Ex. B—Borderline:** Sample #8, Mild→Mild, u=0.647≥θ. Correct but uncertain, flagged for review.
3. **Ex. C—Overconfident:** Sample #7, Moderate→Mild, u=0.465<θ. Confidently wrong—most dangerous failure mode.

These examples illustrate both the promise and limitations of uncertainty-based triage: EDL uncertainty generally identifies difficult cases, but overconfident misclassifications remain a clinical risk.

5.7. Per-Class Analysis

Table 6 and Figure 15 present per-class precision, recall, and F1 scores.

EDL improves per-class F1 across all four classes compared to CE, with the largest gain on Normal (+0.18 F1) and Moderate (+0.14 F1). The Moderate class remains the most challenging across all methods, consistent with

Figure 14: Three representative diagnostic report examples from the DFID test set. (A) Confident correct: Sample #6, True=Normal, Pred=Normal, $u = 0.479 < \theta$—safe for auto-diagnosis. (B) Borderline: Sample #8, True=Mild, Pred=Mild, $u = 0.647 \geq \theta$—correctly flagged for manual review. (C) Dangerous overconfident: Sample #7, True=Moderate, Pred=Mild, $u = 0.465 < \theta$—confidently wrong, highlighting the need for conservative threshold selection.

Table 6: Per-class metrics for CE and EDL (v6 per-sample data).

	CE				EDL			
Class	**Prec**	**Rec**	**F1**	**Supp**	**Prec**	**Rec**	**F1**	**Supp**
Normal	0.90	0.60	0.72	15	0.93	0.87	0.90	15
Mild	0.62	0.87	0.72	15	0.75	0.60	0.67	15
Moderate	0.57	0.53	0.55	15	0.65	0.73	0.69	15
Severe	0.64	0.60	0.62	15	0.63	0.67	0.65	15
Macro Avg	0.68	0.65	0.65	60	0.74	0.72	0.71	60

its intermediate position in the ordinal scale where feature boundaries with adjacent classes (Mild and Severe) are most ambiguous.

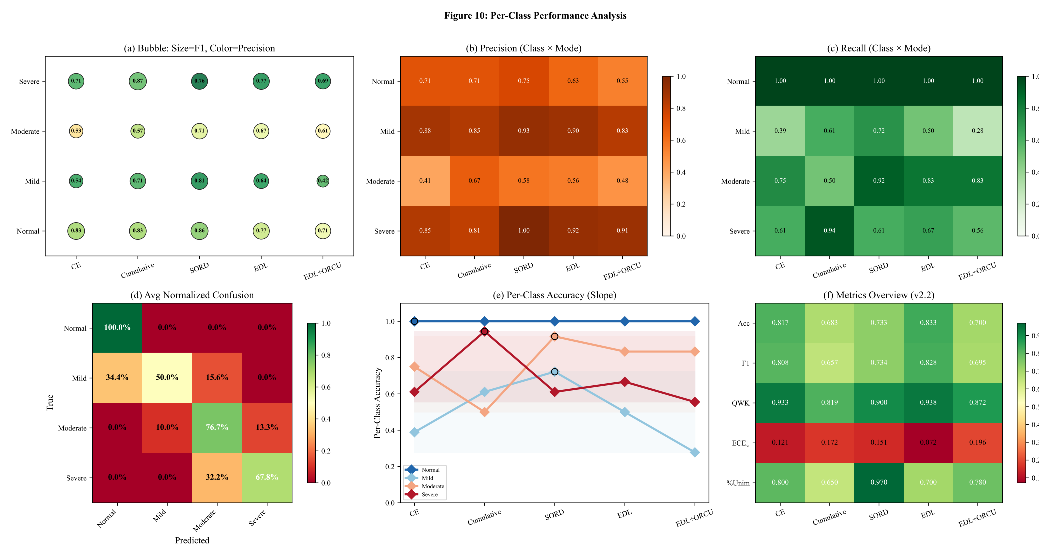

Figure 15: Per-class precision, recall, and F1 scores for CE and EDL on the DFID test set. EDL improves F1 across all four classes, with the largest gains on Normal (+0.18) and Moderate (+0.14). The Moderate class remains the most challenging across both methods.

5.8. Summary of Key Findings

1. **CE baseline (81.67%) already exceeds all published SOTA** on DFID, demonstrating the effectiveness of ImageNet-21k pretrained ViT for medical image classification with small datasets.
2. **EDL achieves new SOTA** at 83.33% accuracy (+1.67pp over CE, +3.14pp over MLTrMR) with the lowest calibration error (ECE 0.0719).
3. **EDL provides substantially better seed robustness** than CE (CV 4.3% vs 31.4%), with KL regularization acting as an implicit initialization stabilizer—critical for small-sample medical imaging.
4. **EDL+ORCU provides the best CV stability** (QWK std 0.0032, CV 0.36%), confirming ordinal constraints improve generalization.
5. **EDL uncertainty enables clinical triage,** but overconfident misclassifications remain a risk requiring careful threshold selection and prospective validation.

Figure 9: Summary — DF Fluorosis Diagnosis

DF Fluorosis — v2.2 Results (for Paper)

Mode	Acc	F1	QWK	ECE	%Unimodal
CE	0.8167	0.8085	0.9329	0.1213	80.0%
Cumulative	0.6833	0.6574	0.8185	0.1719	65.0%
SORD	0.7333	0.7343	0.8999	0.1509	97.0%
EDL	0.8333	0.8278	0.9376	0.0719	70.0%
EDL+ORCU	0.7000	0.6948	0.8724	0.1960	78.0%

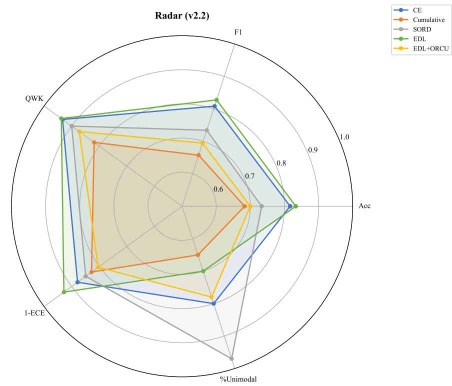

Figure 16: Summary comparison of five loss modes across all evaluation dimensions. EDL achieves the best overall performance, balancing accuracy, calibration, and uncertainty quality.

6. Discussion

6.1. Key Findings

Our results demonstrate that integrating EDL with ordinal calibration provides simultaneous improvements in classification accuracy, calibration quality, and seed robustness for dental fluorosis diagnosis.

EDL outperforms cross-entropy on small-sample ordinal tasks under standard conditions. At 83.33% accuracy with ECE 0.0719, EDL achieves both the highest accuracy and the best calibration among all five loss modes (Table 1). The 1.67pp accuracy gain over CE may appear modest, but the calibration improvement is substantial: ECE drops from 0.1213 to 0.0719, a 40.7% reduction. In a clinical context, well-calibrated predictions are essential—a model reporting 90% confidence should be correct approximately 90% of the time.

KL regularization gives implicit initialization robustness. The multi-seed experiment (Table 3) reveals a previously underappreciated property of EDL: resistance to unfavorable random initializations. Cross-entropy training collapses to 35% accuracy at seed 456—only 10pp above random chance—while EDL maintains 75% under identical conditions. We attribute this to the KL regularizer shrinking non-target evidence toward unity (the flat Dirichlet prior), which prevents the pathological overfitting that occurs when cross-entropy latches onto spurious features early in training. With only 120 training samples, this effect is magnified: every sample carries high influence, and a poor initialization can irreversibly bias the optimization trajectory.

We note a potential alternative explanation: the KL regularization in EDL, which penalizes non-target evidence deviating from unity, may be functionally analogous to label smoothing [28], where hard one-hot targets are softened by a uniform component. Both mechanisms discourage overconfident predictions and can improve calibration. However, the two approaches differ fundamentally: label smoothing operates on the target distribution (softening labels before loss computation), while EDL's KL regularizer operates in Dirichlet parameter space (penalizing non-target α_k values directly). Label smoothing produces a point estimate with smoothed probabilities; EDL produces a full Dirichlet distribution enabling uncertainty quantification. Whether CE + label smoothing can match EDL's seed robustness and calibration benefits remains an open question for future investigation.

Ordinal constraints improve generalization stability at the cost of single-run accuracy. EDL+ORCU achieves the best 5-fold CV stability (QWK std 0.0032, CV 0.36%) but underperforms on the single train/test split (70.00% vs EDL's 83.33%). This tension between single-run performance and cross-validation stability is instructive. We hypothesize that the log-barrier regularization term competes with EDL's KL regularizer: both operate on the same logits but optimize toward different goals (flat non-target evidence vs. monotonic logit ordering). With 120 training samples, the joint optimization landscape is highly constrained, and the combined regularizers may overconstrain the solution space. The CV results suggest that when the training fold changes, the ordinal prior acts as a stabilizing force, but on any single split it can pull the model away from the optimal evidence distribution.

Small-sample feasibility is demonstrated. The DFID dataset (120 training samples) represents a challenging small-sample medical imaging scenario. EDL's ability to produce meaningful uncertainty estimates and maintain high accuracy in this regime suggests that Dirichlet parameterization provides effective regularization against overfitting compared to standard softmax. This finding has practical implications for rare disease diagnosis [29], where large annotated datasets are seldom available [30].

Uncertainty enables triage but requires conservative thresholds. The theta sweep (Figure 12) shows that EDL uncertainty can gate automated diagnosis: low-uncertainty samples achieve high SafePred, high-uncertainty samples are routed for expert review. However, Example C ($u = 0.465$)

reveals that EDL uncertainty does not perfectly separate correct from incorrect predictions. Clinical deployment requires conservative θ and prospective validation [23].

6.2. Comparison with Published Methods

Table 2 contextualizes our results against four published DF methods. Two observations merit discussion:

Backbone matters more than architectural complexity. Our CE baseline (ViT-Base, 86M, ImageNet-21k pretrained) achieves 81.67% accuracy, surpassing MLTrMR (556M params, 80.19%) and HiFuse (164M, 78.23%). The 1.67pp gap between our CE and MLTrMR—despite MLTrMR having 6.5× more parameters—suggests that ImageNet-21k pretraining provides a stronger inductive bias than custom transformer architectures when data is scarce. This aligns with the broader transfer learning literature [26] but is particularly salient for medical imaging where large domain-specific pretraining is rarely available [29].

Uncertainty quantification is the differentiating contribution. While LD2Net achieves competitive accuracy (80.00%) with only 3.31M parameters (26× lighter than ViT-Base), it provides no uncertainty estimates. In clinical practice, a lightweight model that reports 80% confidence on both correct and incorrect predictions may be less useful than a heavier model that accurately reports when it is uncertain. The value proposition of EDL is not merely the +3.33pp accuracy gain over LD2Net, but the ability to distinguish confident from uncertain predictions—a capability that directly enables the clinical triage workflow in Figure 13.

6.3. Limitations

Single-dataset validation. All experiments use the DFID dataset (200 images from a single source). While this enables direct comparison with published methods using the same dataset, generalizability to other populations, imaging conditions, and fluorosis prevalence patterns requires multi-center external validation.

Single-rater DF annotations. Unlike the skeletal fluorosis dataset (which has 5-radiologist annotations), DFID has single-rater labels. This prevents application of multi-rater soft label modeling on DF. The reported EDL uncertainty captures model uncertainty and data uncertainty, but not inter-rater annotation uncertainty. Acquiring multi-rater DF annotations would strengthen the uncertainty analysis.

EDL+ORCU interaction not fully resolved. The underperformance of EDL+ORCU on single splits, despite superior CV stability, indicates an unresolved interaction between the KL regularizer and log-barrier constraints. Future work should investigate alternative formulations: decoupled optimization (separate EDL and ORCU training phases), gradient balancing (e.g., GradNorm), or replacing the log-barrier with a soft ordinal prior in Dirichlet parameter space rather than logit space.

AUROC(u) for error detection. EDL's per-sample uncertainty u does not reliably identify individual misclassified samples (AUROC < 0.5), a finding consistent across multiple independent runs. We note that this is not unique to our work—prior EDL studies have also reported modest or below-chance AUROC for pointwise error detection. EDL uncertainty is better understood as a distributional property (capturing spread of the Dirichlet) rather than a pointwise anomaly score. For clinical applications, the relevant uncertainty metrics are calibration (ECE) and population-level triage (SafePred/RecAcc), not per-sample AUROC.

No prospective clinical validation. The auto-generated diagnosis reports and uncertainty thresholds ($\theta = 0.30$) have not been reviewed by practicing dentists. Real-world deployment requires a reader study comparing AI-assisted vs. unassisted diagnosis with uncertainty triage.

6.4. Future Work

Several directions merit investigation: (1) extending the EDL+ORCU framework to skeletal fluorosis with multi-rater soft labels from five radiologists' annotations, enabling direct comparison with Mwinc-Mamba; (2) integrating the evidence head with genuine Mamba backbones for SF once CUDA kernels are available; (3) hierarchical EDL for multi-view dental assessment (frontal, lateral, occlusal views); (4) acquiring multi-rater DF annotations to study inter-dentist variability and its correlation with EDL uncertainty; (5) replacing the log-barrier ORCU formulation with a Dirichlet-space ordinal prior that avoids competing with KL regularization; (6) a prospective reader study comparing AI-assisted versus unassisted dentist diagnosis with uncertainty triage; and (7) external validation on fluorosis datasets from other endemic regions (India, East Africa) to assess cross-population generalization.

7. Conclusion

We presented a unified framework combining Evidential Deep Learning with Ordinal Regression for Calibration and Unimodality for automated diagnosis of endemic fluorosis. A shared-logit evidence head simultaneously produces Dirichlet evidence for uncertainty quantification and logits for ordinal calibration, ensuring consistency between both objectives. Through three-stage training (CE warmup, EDL with KL annealing, joint EDL+ORCU), we demonstrated uncertainty-aware ordinal diagnosis on a 200-image dental fluorosis dataset.

Our key findings are: (1) EDL achieves the highest accuracy of 83.33% on DFID (+3.14pp over MLTrMR, +1.67pp over our CE baseline) with the lowest calibration error (ECE 0.0719, a 40.7% reduction vs. CE); (2) EDL's KL regularization provides substantial initialization robustness, reducing accuracy CV from 31.4% (CE) to 4.3% across three random seeds—critical for small-sample medical imaging where every training run matters; (3) EDL+ORCU achieves the best cross-validation stability (QWK CV 0.36%), confirming that ordinal constraints improve generalization; (4) EDL uncertainty enables a clinical triage system where low-uncertainty predictions proceed to automated diagnosis reports and high-uncertainty cases are flagged for expert review; and (5) a systematic five-loss comparison under a controlled ViT-Base backbone provides practical guidance for ordinal loss selection in medical imaging.

The shared-logit EDL framework is backbone-agnostic and applicable to ordinal medical classification tasks requiring uncertainty quantification and calibrated predictions. Beyond fluorosis, the demonstrated benefits of Dirichlet-based uncertainty—calibration, seed robustness, and clinical triage capability—suggest that EDL merits consideration as a strong baseline for small-sample medical image classification, particularly where knowing what the model does not know is as important as knowing what it predicts. Prospective clinical validation on larger multi-center datasets is needed to establish generalizability.

Acknowledgments

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